OPERATIONS RESEARCH
Vol. 61, No. 3, May–June 2013, pp. 677–693
ISSN 0030-364X (print) | ISSN 1526-5463 (online)

http://dx.doi.org/10.1287/opre.1120.1136
© 2013 INFORMS

The Robust Capacitated Vehicle Routing Problem Under Demand Uncertainty

Chrysanthos E. Gounaris
Department of Chemical and Biological Engineering, Princeton University, Princeton, New Jersey 08544, gounaris@princeton.edu

Wolfram Wiesemann
Imperial College Business School, London SW7 2AZ, United Kingdom, ww@imperial.ac.uk

Christodoulos A. Floudas
Department of Chemical and Biological Engineering, Princeton University, Princeton, New Jersey 08544, floudas@princeton.edu

The robust capacitated vehicle routing problem (CVRP) under demand uncertainty is studied to address the minimum cost delivery of a product to geographically dispersed customers using capacity-constrained vehicles. Contrary to the deterministic CVRP, which postulates that the customer demands for the product are deterministic and known, the robust CVRP models the customer demands as random variables, and it determines a minimum cost delivery plan that is feasible for all anticipated demand realizations. Robust optimization counterparts of several deterministic CVRP formulations are derived and compared numerically. Robust rounded capacity inequalities are developed, and it is shown how they can be separated efficiently for two broad classes of demand supports. Finally, it is analyzed how the robust CVRP relates to the chance-constrained CVRP, which allows a controlled degree of supply shortfall to decrease delivery costs.

Subject classifications: programming: robust optimization; transportation: vehicle routing; probability: chance constraints.
Area of review: Optimization.
History: Received August 2011; revisions received March 2012, August 2012; accepted September 2012. Published online in *Articles in Advance* February 5, 2013.

1. Introduction

Vehicle routing problems have been investigated since the 1950s, and they constitute some of the best-studied problems in operations research. They have applications in domains as diverse as courier delivery, waste collection, dial-a-ride services, as well as the routing of school buses, snow plow trucks, and maintenance engineers. One of the most commonly adopted vehicle routing formulations is the capacitated vehicle routing problem (CVRP), which concerns the minimum cost delivery of a single product from a depot to customers through a specified number of capacity-constrained vehicles. For recent surveys on the CVRP and its variants, the reader is directed to Baldacci et al. (2012, 2010), Cordeau et al. (2007), Golden et al. (2008), Laporte (2009), and Toth and Vigo (2002). The classical CVRP is a deterministic optimization problem because it is assumed that the customer demands and the transportation costs are known with certainty. In this paper, we study the robust CVRP under demand uncertainty.

Stochastic programming models the uncertain parameters of an optimization problem as random variables with known probability distributions. The two basic types of stochastic programs are recourse problems and chance-constrained problems. In a two-stage recourse problem, the decision maker takes a here-and-now decision (taken before the realizations of the uncertain parameters are known) and a recourse decision (selected after the realizations of the uncertain parameters are revealed) that jointly optimize a risk functional—for example, the expected value. In a chance-constrained problem, on the other hand, the decision maker selects a here-and-now decision that satisfies all constraints with a pre-specified probability. The stochastic programming literature is reviewed in Birge and Louveaux (2011), Prékopa (1995), and Shapiro et al. (2009).

In the last four decades, various recourse and chance-constrained stochastic programs have been developed for the stochastic CVRP. It turns out that the stochastic CVRP is considerably more challenging than its deterministic counterpart because optimal solutions lack the structural properties known from the deterministic case (Dror et al. 1989, Jaillet 1988). The recourse formulations model the vehicle routes as here-and-now decisions that are chosen before the uncertain customer demands are realized, whereas the recourse decisions are taken once the customer demands are observed. Possible recourse decisions include penalties that are incurred for each unit of unsatisfied demand (Stewart and Golden 1983), detours to the depot that are taken whenever a customer's demand exceeds the remaining vehicle load (Bertsimas and Simchi-Levi 1996, Dror and Trudeau 1986), the decision to serve customers by other vehicles (Ak and Erera 2007), and the option to preventively restock a vehicle during its route (Yang

et al. 2000). Most of the exact solution techniques for the stochastic CVRP with recourse rely on variants of the integer L-Shaped method (Laporte et al. 2002) or branch-and-price algorithms (Christiansen et al. 2009).

The chance-constrained CVRP considers here-and-now vehicle routes that satisfy the customer demands along each route with a pre-specified probability. This problem is typically solved using branch-and-cut methods (Laporte et al. 1992), but in special cases, such as independent and identically distributed demands, the problem can be reformulated as a deterministic vehicle routing problem (Dror et al. 1993, Golden and Yee 1979). For reviews of the stochastic CVRP, see Cordeau et al. (2007), Gendreau et al. (1996), and Toth and Vigo (2002). Apart from the stochastic CVRP with uncertain demands, there is a large body of literature on vehicle routing problems with uncertain customers (Bent and Hentenryck 2004, Gendreau et al. 1995) and uncertain travel times (Kenyon and Morton 2003), dynamic problems in which customer requests arrive over time (Secomandi and Margot 2009, Smith et al. 2010), and stochastic inventory routing problems that combine vehicle routing with inventory control (Adelman 2004, Kleywegt et al. 2004).

Similar to stochastic programming, robust optimization models the uncertain parameters of an optimization model as random variables. The early literature on robust optimization aims to determine minimum cost here-and-now decisions that remain feasible for any realization of the parameters over some prespecified support, such as budget sets, polyhedra, or intersections of ellipsoids. Under this assumption, robust counterparts have been developed for various types of convex optimization problems (Ben-Tal et al. 2009, Bertsimas and Sim 2004), as well as combinatorial and mixed-integer problems (Bertsimas and Sim 2003; Janak et al. 2007; Li et al. 2011; Lin et al. 2004; Verderame et al. 2010; Verderame and Floudas 2009, 2011). More recent contributions account for additional information about the probability distribution governing the random variables (Chen et al. 2010, Delage and Ye 2010, El Ghaoui et al. 2003). Moreover, robust optimization has been extended to multi-stage problems where the here-and-now decision is amended by a recourse decision (Ben-Tal et al. 2009, 2004; Chen et al. 2008; Erera et al. 2009; Kuhn et al. 2011). For reviews of the robust optimization literature, we refer to Ben-Tal et al. (2009) and Bertsimas et al. (2011).

To our knowledge, the first robust CVRP has been proposed in Sungur et al. (2008). In this paper, it is assumed that the customer demands and travel times are uncertain. The goal is to determine vehicle routes that satisfy the vehicle capacities and specified delivery time windows if all customer demands and travel times can attain their worst-case realizations simultaneously. Under this assumption, the problem simplifies to a deterministic CVRP, which is very attractive from a computational viewpoint. The authors apply variants of the model to a bioterrorism emergency planning problem in Shen et al. (2009) and a courier delivery problem in Sungur et al. (2010), respectively.

The robust CVRP presented in Sungur et al. (2008) can lead to conservative solutions because it anticipates that each customer demand and travel time can attain its worst-case realization simultaneously. Through clever problem reformulations and the introduction of auxiliary binary variables, Ordóñez (2010) generalize the model in Sungur et al. (2008) to allow for upper bounds on the customer demands and travel times experienced by each vehicle.

A hybrid model that combines stochastic programming and robust optimization is presented in Erera et al. (2010). The authors elaborate a two-stage recourse problem in which here-and-now vehicle routes are selected before the uncertain customer demands are realized. Once the demands are observed, a recourse decision is taken that amends the routes with returns to the depot whenever a vehicle needs to be replenished. The model minimizes the sum of expected route durations, subject to upper bounds on the route durations that must be satisfied for all anticipated demand realizations. The authors develop a Tabu search heuristic to solve this problem.

Finally, a multi-stage robust optimization problem for humanitarian relief logistics planning is presented in Ben-Tal et al. (2011). In this problem, an uncertain amount of traffic is routed through a network at minimum costs such that specified road capacities are obeyed. Contrary to the robust CVRP, traffic is modeled by continuous variables, and there are no vehicle capacities. The resulting problem can be solved as a linear program.

In this paper we study the formulation and solution of the robust CVRP under demand uncertainty. Contrary to the existing contributions to this problem, we consider the generic case where the customer demands are supported on a (typically nonrectangular) polyhedron. This avoids overly conservative solutions that hedge against the unlikely scenario where all customer demands attain their worst-case realizations simultaneously. It also enables us to inject statistical information into the problem and relate the robust CVRP to the chance-constrained CVRP. The resulting problem does not reduce to a deterministic CVRP, and it necessitates some reformulation to make it amenable to numerical solution. We can summarize the contributions of this paper as follows.

1. We show conditions that allow the robust CVRP to reduce to the solution of a deterministic CVRP. This is the case if the support of the customer demands is rectangular, or if it can be represented as the intersection of a hyper-rectangle with a single budget constraint.

2. For supports that do not satisfy the aforementioned properties, we present necessary and sufficient conditions for the correctness of the robust CVRP. We then develop the robust counterparts of several known deterministic CVRP formulations: the two-index vehicle flow, the Miller-Tucker-Zemlin, as well as the one-commodity flow and two-commodity flow formulations. We prove that each of

the robust counterparts satisfies the necessary and sufficient conditions, and we derive finite-dimensional reformulations that are amenable to numerical solution schemes.

3. We present a new formulation for the deterministic CVRP—the vehicle assignment formulation—that is reminiscent of the three-index vehicle flow formulation but requires fewer binary variables. We derive the robust counterpart of this formulation, and we prove that the robust counterpart also satisfies the necessary and sufficient conditions. We argue that the robust vehicle assignment formulation is particularly suited from a computational viewpoint.

4. We derive the robust counterpart of the rounded capacity inequalities (RCI). In general, separating these inequalities in a branch-and-cut method requires the repeated solution of linear programs. We show that a much more efficient separation is possible under mild assumptions regarding the support of the uncertain customer demands, and we demonstrate that this separation expedites significantly the solution of the robust CVRP.

5. We establish the connection between the robust CVRP and a distributionally robust variant of the chance-constrained CVRP. While relationships of this type are known to the robust optimization community, they seem to be new in the context of the CVRP.

Our results bear new and insightful connections to the optimality of affine decision rules in multistage robust optimization problems, which we will emphasize throughout the text.

The remainder of the paper is structured as follows. Section 2 defines the robust CVRP with demand uncertainty. In §3, we demonstrate that the robust CVRP reduces to the solution of a deterministic CVRP if the support of the uncertain customer demands satisfies certain "rectangularity" properties. We derive several equivalent formulations for the robust CVRP with generic (nonrectangular) support in §4. Section 5 shows how robust RCI cuts can be separated during the solution of robust CVRP instances. Section 6 provides computational results, and we conclude in §7. An electronic companion to this paper is available as part of the online version at http://dx.doi.org/10.1287/opre.1120.1136. In the electronic companion of this paper, we discuss how the robust CVRP relates to the chance-constrained CVRP, which allows for a controlled degree of supply shortfall to reduce delivery costs. For readability, we also defer all proofs to this e-companion.

Notation. We use the indicator function $\mathbb{1}_{[\mathscr{E}]}$ to denote 1 if the logical expression $\mathscr{E}$ is true and 0 otherwise. The column vector e denotes a vector of ones; its dimensionality will become clear from the context. For a vector x, $\|x\|_p$ denotes the p-norm of x, and we define $[x]^+ = \max\{x, 0\}$.

2. Problem Definition

The deterministic CVRP can be described as follows. Let $G = (V, A)$ be a complete directed graph with nodes $V = \{0, 1, \ldots, n\}$ and arcs $A = \{(i, j) \in V \times V\colon i \neq j\}$. Node $0 \in V$ represents the unique depot, and each node $i \in V\backslash\{0\}$ corresponds to a customer with demand $q_i \in \mathbb{R}_+$. For notational convenience, we define $V_C = V\backslash\{0\}$ as the set of customer nodes. The depot is equipped with m homogeneous vehicles of capacity Q. We denote the set of vehicles as $K = \{1, \ldots, m\}$. Each vehicle incurs a transportation cost $c(i, j) \in \mathbb{R}_+$ if it traverses the arc $(i, j) \in A$.

A *set of routes* $\mathbf{R} = (R_1, \ldots, R_m)$ to the CVRP represents a partition of the customer set V_C into m routes $R_k = (R_{k,1}, \ldots, R_{k,n_k})$, where $R_{k,l}$ represents the lth customer and n_k the number of customers visited by vehicle k, respectively. We use the notation $i \in R_k$ to indicate that customer i is served by vehicle k, that is, $i = R_{k,l}$ for some $l \in \{1, \ldots, n_k\}$. A set of routes $\mathbf{R}$ is *feasible* if $\sum_{i \in R_k} q_i \leqslant Q$ for all $k \in K$, that is, if it satisfies the capacity constraints of all vehicles. We denote by $\mathscr{R}(q)$ the set of feasible route sets to the CVRP for a given demand vector q. The goal of the CVRP is to determine a feasible set of routes that minimizes the cumulative transportation cost:

$$\begin{aligned} &\underset{\mathbf{R}}{\text{minimize}} \quad c(\mathbf{R}) = \sum_{k \in K} \sum_{l=0}^{n_k} c(R_{k,l}, R_{k,l+1}), \\ &\text{subject to} \quad \mathbf{R} \in \mathscr{R}(q), \end{aligned} \tag{$\mathscr{CVRP}$}$$

where we define $R_{k,0} = R_{k,n_k+1} = 0$ for all $k \in K$, that is, each route starts and ends at the depot. Because $\mathbf{R}$ represents a partition of the customer set V_C, all customers must be served by a vehicle, and split deliveries are not allowed. Moreover, subtours are excluded because each route has to start and end at the depot. In the remainder of the paper, we abbreviate $c(i, j)$ by c_{ij} for $(i, j) \in A$.

In practice, the customer demands are often subject to significant uncertainty. In the following, we model the customer demands as random variables and assume that only the support $\mathscr{Q} \subseteq \mathbb{R}^n_+$ of the demands is known. In this case, we call a set of routes $\mathbf{R}$ *robust feasible* if $\sum_{i \in R_k} q_i \leqslant Q$ for all $k \in K$ and $q \in \mathscr{Q}$, that is, if $\mathbf{R}$ satisfies the capacity constraints of all vehicles for all anticipated demand realizations. Let $\mathscr{R}(\mathscr{Q})$ be the set of robust feasible route sets for a given support $\mathscr{Q}$, that is, $\mathscr{R}(\mathscr{Q}) = \bigcap_{q \in \mathscr{Q}} \mathscr{R}(q)$. Throughout the paper, we assume that $\mathscr{Q}$ satisfies the following conditions:

(C1) The support $\mathscr{Q}$ is a nonempty and closed subset of $\mathbb{R}^n_+$.

(C2) For each customer $i \in V_C$, we have $\max\{q_i\colon q \in \mathscr{Q}\} > 0$.

Condition (C1) stipulates that demands are nonnegative. Condition (C2) can always be ensured by removing redundant customers. We sometimes also assume that $\mathscr{Q}$ has a particular shape:

(C3) The support $\mathscr{Q}$ is polyhedral: $\mathscr{Q} = \{q \in \mathbb{R}^n_+\colon Wq \leqslant h\}$, where $W \in \mathbb{R}^{r \times n}$ and $h \in \mathbb{R}^r$.

Our formulations extend to the case $r = 0$, where the customer demands are supported on $\mathbb{R}^n_+$. Although uninteresting for the robust CVRP, this case becomes meaningful when we study the chance-constrained problem. Our results can also be extended to supports $\mathscr{Q}$ that constitute intersections of (possibly degenerate) ellipsoids. For the sake of brevity, we disregard this generalization.

In the robust CVRP, the decision maker selects a robust feasible set of routes that minimizes the overall transportation cost:

$$\begin{aligned} &\underset{\mathbf{R}}{\text{minimize}} \quad c(\mathbf{R}), \\ &\text{subject to} \quad \mathbf{R} \in \mathcal{R}(\mathcal{Q}). \end{aligned} \tag{$\mathcal{RVRP}$}$$

Thus, the goal is to determine here-and-now vehicle routes that satisfy the vehicle capacity restrictions for all anticipated realizations of the customer demands. Although no recourse decision is taken in $\mathcal{RVRP}$, all our results extend to simple recourse decisions that penalize each unit of supply shortfall. Clearly, $\mathcal{RVRP}$ might become overly risk-averse or even infeasible if $\mathcal{Q}$ is chosen to be very large (e.g., all theoretically possible demand realizations). This can be avoided, for example, by choosing $\mathcal{Q}$ to be the smallest set that contains all historically observed demand realizations (possibly disregarding some outliers). Alternatively, by injecting statistical information about the demand distribution, we can obtain less conservative solutions that trade off route feasibility for savings in transportation costs (see §EC.3 in the paper's e-companion). Even if historical information is not available, a subjective choice of $\mathcal{Q}$ based on expert opinions can successfully immunize the routing decision against estimation errors in the demands. Indeed, it has been shown in Ben-Tal et al. (2009) that the robustness of a solution can often be increased significantly at a minor deterioration of the objective value.

Note that $\mathcal{RVRP}$ is *not* equivalent to the worst-case deterministic CVRP, which we define as

$$\max_{q \in \mathcal{Q}} \left\{ \begin{array}{ll} \underset{\mathbf{R}}{\text{minimize}} & c(\mathbf{R}) \\ \text{subject to} & \mathbf{R} \in \mathcal{R}(q) \end{array} \right\}. \tag{$\overline{\mathcal{CVRP}}$}$$

Intuitively, $\overline{\mathcal{CVRP}}$ determines a maximum cost (deterministic) CVRP instance in $\mathcal{Q}$, whereas $\mathcal{RVRP}$ determines a minimum cost solution that is feasible for all anticipated demand realizations $q \in \mathcal{Q}$ simultaneously. In other words, $\overline{\mathcal{CVRP}}$ allows the routes $\mathbf{R}$ to be chosen *after* the demand realization $q \in \mathcal{Q}$ is observed, whereas $\mathcal{RVRP}$ requires the routes to be selected *before* the demand realization is known.

3. Relationship to the Worst-Case Deterministic CVRP

In this section we examine properties of the support $\mathcal{Q}$ of the uncertain customer demands that allow us to determine an optimal solution to the robust problem $\mathcal{RVRP}$ by solving an instance of the deterministic problem $\mathcal{CVRP}$. To this end, let us first consider the case where the support $\mathcal{Q}$ is *rectangular*, that is, where

$$\mathcal{Q} = \{q_1 \colon q \in \mathcal{Q}\} \times \cdots \times \{q_n \colon q \in \mathcal{Q}\}. \tag{1}$$

For a rectangular support, $\mathcal{RVRP}$ determines a set of routes that remains feasible if each customer demand attains its worst realization individually, irrespective of the other customer demands. Note that the mapping $\mathcal{R}$ of feasible route sets is monotonous, that is, $\mathcal{R}(q') \subseteq \mathcal{R}(q)$ for $q' \geqslant q$. Thus, solving $\mathcal{RVRP}$ over a rectangular support $\mathcal{Q}$ is equivalent to solving $\mathcal{CVRP}$ for the worst-case demand vector $\bar{q} \in \mathcal{Q}$, where $\bar{q}$ is defined through $\bar{q}_i = \max\{q_i \colon q \in \mathcal{Q}\}$, $i \in V_C$.

If the support $\mathcal{Q}$ is not rectangular, then we can define its *rectangular projection* $\bar{\mathcal{Q}} \supset \mathcal{Q}$ as the right-hand side of the identity in (1). It is proposed in Sungur et al. (2008) to solve $\mathcal{RVRP}$ by replacing $\mathcal{Q}$ with $\bar{\mathcal{Q}}$. From the previous argumentation we know that this is equivalent to solving $\mathcal{CVRP}$ for the demand vector $\bar{q}$. In the following, we denote this problem as $\overline{\mathcal{RVRP}}$. We now show that $\overline{\mathcal{RVRP}}$ provides a conservative approximation to $\mathcal{RVRP}$.

Proposition 1. *For any support $\mathcal{Q}$, we have*

$$\overline{\mathcal{CVRP}} \preceq \mathcal{RVRP} \preceq \overline{\mathcal{RVRP}}, \tag{2}$$

where "$\preceq$" refers to the ordering of optimal objective values between the problems. Moreover, the optimal objective values of the three problems coincide if $\mathcal{Q}$ is rectangular.

Section EC.1 in the paper's e-companion shows that the inequalities in (2) are strict in general. Proposition 1 implies that for rectangular supports $\mathcal{Q}$, we obtain an optimal solution to $\mathcal{RVRP}$ by solving $\mathcal{CVRP}$ with $q = \bar{q}$, where $\bar{q}$ is defined through $\bar{q}_i = \max\{q_i \colon q \in \mathcal{Q}\}$, $i \in V_C$. In fact, we can obtain an optimal solution to $\mathcal{RVRP}$ through the solution of a single instance of $\mathcal{CVRP}$ in a more general setting.

Proposition 2. *For supports of the form $\mathcal{Q} = \{q \in [0, \bar{q}] \colon \sum_{i \in V_C} q_i \leqslant B\}$, $\mathcal{RVRP}$ reduces to the solution of a single instance of $\mathcal{CVRP}$.*

The supports in Proposition 2 result from the intersection of the hyper-rectangle $[0, \bar{q}]$ with a single halfspace that imposes an upper bound on the cumulative customer demands. These supports can be interpreted as special cases of the budget uncertainty sets presented in Bertsimas and Sim (2003, 2004). Translated into our context, the budget uncertainty sets in these papers additionally allow for lower bounds $\underline{q}$ on the uncertain customer demands. It seems unlikely that the proposition can be generalized significantly. Indeed, the support in §EC.1 deviates from the assumptions in Proposition 2 only through the omission of some customer demands in the inequality $\sum_{i \in V_C} q_i \leqslant B$, and it is not obvious how the associated instance of $\mathcal{RVRP}$ would reduce to the solution of an instance of $\mathcal{CVRP}$. Similarly, one can construct $\mathcal{RVRP}$ instances with lower bounds $\underline{q}$ that do not seem to reduce to the solution of an instance of $\mathcal{CVRP}$.

4. Robust Counterparts of Deterministic CVRP Formulations

Section EC.1 illustrates that the optimal solutions to $\mathcal{RVRP}$ and $\overline{\mathcal{RVRP}}$ differ in general. Indeed, one can

easily modify the example presented there to show that replacing the support $\mathcal{Q}$ with its rectangular projection $\bar{\mathcal{Q}}$ can result in arbitrarily poor approximations of $\mathcal{RVRP}$. Therefore, this section considers the generic case where the support $\mathcal{Q}$ does not satisfy the rectangularity condition (1).

So far, we have characterized solutions to $\mathcal{CVRP}$ and $\mathcal{RVRP}$ by their route sets $\mathbf{R} = (R_1, \ldots, R_m)$. In practice, however, mathematical models for $\mathcal{CVRP}$ describe solutions through binary variables x_{ij} that record whether a vehicle traverses the arc $(i, j) \in A$:

$$x_{ij} = 1 \iff \exists k \in K, \quad 0 \leqslant l \leqslant n_k\colon (i, j) = (R_{k,l}, R_{k,l+1}). \tag{3}$$

By construction, any set of routes $\mathbf{R}$ to $\mathcal{CVRP}$ or $\mathcal{RVRP}$ induces unique values for x_{ij} through this relationship. The following observation reiterates the well-known fact that under certain conditions, the values of x_{ij} also induce a unique set of routes $\mathbf{R}$ (Cordeau et al. 2007, Golden et al. 2008, Toth and Vigo 2002). For brevity, we omit the proof of the statement.

Observation 1. The binary variables x_{ij}, $(i, j) \in A$ induce a set of m routes $\mathbf{R}$ through the equivalence (3) if and only if x_{ij} satisfy the following requirements:

(i) *Degree restrictions.* Each customer $i \in V_C$ is visited by exactly one vehicle, and m vehicles depart from and return to the depot:

$$\sum_{\substack{j \in V:\\ (i,j) \in A}} x_{ij} = \sum_{\substack{j \in V:\\ (j,i) \in A}} x_{ji} = \delta_i \quad \forall i \in V, \tag{4}$$

where $\delta_i = 1$ if $i \in V_C$ and $\delta_0 = m$.

(ii) *Subtour restrictions.* Each customer $i \in V_C$ is visited by a vehicle $k \in K$ that departs from and returns to the depot:

$$\nexists \{i_1, \ldots, i_s\} \subseteq V_C\colon \quad x_{i_1,i_2} = x_{i_2,i_3} = \cdots = x_{i_{s-1},i_s} = x_{i_s,i_1} = 1. \tag{5}$$

The set of routes $\mathbf{R}$ induced by x_{ij} is unique up to a reordering of the individual routes $R_1, \ldots, R_m$.

Whenever the requirements of Observation 1 are satisfied, we denote by $\mathbf{R}(x)$ the unique set of routes $\mathbf{R}$ that is induced by x. The binary variables x_{ij} induce a feasible set of routes $\mathbf{R}(x)$, that is, a feasible solution to the deterministic problem $\mathcal{CVRP}$, if and only if x_{ij} satisfy the requirements in Observation 1 and $\sum_{i \in R_k(x)} q_i \leqslant Q$ for all $k \in K$. Likewise, x_{ij} induce a robust feasible set of routes $\mathbf{R}(x)$, that is, a feasible solution to the robust problem $\mathcal{RVRP}$, if and only if x_{ij} satisfy the requirements in Observation 1 and $\sum_{i \in R_k(x)} q_i \leqslant Q$ for all $k \in K$ and $q \in \mathcal{Q}$.

In this section, we review various deterministic CVRP formulations that impose constraints that are necessary and sufficient for x to induce a feasible solution $\mathbf{R}(x)$ to $\mathcal{CVRP}$. For each of these formulations, we derive a robust counterpart that imposes constraints that are necessary and sufficient for x to induce a feasible solution $\mathbf{R}(x)$ to $\mathcal{RVRP}$. In particular, we develop the robust counterparts of the two-index vehicle flow formulation (§4.1), the Miller-Tucker-Zemlin formulation (§4.2), and the commodity flow formulations (§4.3). We also present a new vehicle assignment formulation in §4.4. Section EC.2 in the paper's e-companion compares the numbers of variables and constraints required for each of these formulations.

4.1. Two-Index Vehicle Flow Formulation

The two-index vehicle flow formulation is based on the rounded capacity inequalities (RCI), which constitute a generalization of the generalized subtour elimination constraints developed for the traveling salesman problem. Both the deterministic and the robust RCI constraints enforce lower bounds on the number of vehicles that serve a customer set $S \subseteq V_C$,

$$\sum_{i \in V \setminus S} \sum_{j \in S} x_{ij}.$$

In the deterministic CVRP, this is achieved through the *deterministic RCI constraints*

$$\sum_{i \in V \setminus S} \sum_{j \in S} x_{ij} \geqslant \left\lceil \frac{1}{Q} \sum_{i \in S} q_i \right\rceil \quad \forall S \subseteq V_C; \tag{6}$$

see Laporte et al. (1985). If the customer demands are strictly positive, then the deterministic RCI constraints require that every customer subset $S \subseteq V_C$ is visited by at least one vehicle that enters S from a node $i \in V \setminus S$. Together with the degree constraints (4), the deterministic RCI constraints therefore exclude subtours, and Observation 1 implies that x induces a set of routes $\mathbf{R}(x)$. Moreover, for any route $R_k(x)$, the left-hand side of the deterministic RCI constraint associated with $S = R_k(x)$ evaluates to one, which implies satisfaction of the vehicles' capacity constraints, that is, $\mathbf{R}(x)$ is feasible in $\mathcal{CVRP}$.

In the following, we show that in the robust CVRP, tight lower bounds on the number of vehicles that need to serve customer sets $S \subseteq V_C$ are obtained from the following *robust RCI constraints*:

$$\sum_{i \in V \setminus S} \sum_{j \in S} x_{ij} \geqslant \left\lceil \frac{1}{Q} \max_{q \in \mathcal{Q}} \sum_{i \in S} q_i \right\rceil \quad \forall S \subseteq V_C. \tag{7}$$

Proposition 3. *For any support $\mathcal{Q}$ and binary variables x_{ij}, $(i, j) \in A$, we have that*

(i) *the robust RCI constraints* (7) *are* necessary *to induce a robust feasible set of routes* $\mathbf{R}(x)$;

(ii) *in conjunction with the degree constraints* (4), *the robust RCI constraints* (7) *are* sufficient *to induce a robust feasible set of routes* $\mathbf{R}(x)$.

Note that the right-hand side of each robust RCI constraint (7) does not depend on x. Hence, under condition (C3) we can replace this right-hand side with the optimal objective value of a linear program. The $\mathcal{RVRP}$ with robust RCI constraints is described in Formulation (2VF).

Formulation (2VF). Robust two-index vehicle flow formulation.

$$\begin{aligned}
\underset{x}{\text{minimize}} \quad & \sum_{(i,j)\in A} c_{ij}x_{ij}, \\
\text{subject to} \quad & x_{ij} \in \{0,1\},\ (i,j)\in A \\
& \sum_{\substack{j\in V:\\ (i,j)\in A}} x_{ij} = \sum_{\substack{j\in V,\\ (j,i)\in A}} x_{ji} = \delta_i \quad \forall i \in V, \\
& \sum_{i\in V\setminus S}\sum_{j\in S} x_{ij} \geqslant \left\lceil \frac{1}{Q}\max_{q\in\mathcal{Q}}\sum_{i\in S} q_i \right\rceil \quad \forall S \subseteq V_C.
\end{aligned}$$

4.2. Miller-Tucker-Zemlin Formulations

We first derive the robust counterpart of the classical Miller-Tucker-Zemlin (MTZ) constraints, which were presented for the deterministic CVRP in Kulkarni and Bhave (1985). In §4.2.2, we develop a more efficient robust counterpart that exploits our structural knowledge about the optimal solution to the robust MTZ formulation.

4.2.1. Robust Miller-Tucker-Zemlin Formulation. Both the deterministic and the robust MTZ constraints encode the cumulative customer demands served by the vehicle that visits customer $i \in V_C$, immediately after it has left customer i. In the deterministic case, this is achieved through the *deterministic MTZ constraints* (Kulkarni and Bhave 1985):

$$\begin{aligned}
& u_j - u_i + Q(1-x_{ij}) \geqslant q_j \quad \forall i, j \in V_C,\ i \neq j, \\
& q_i \leqslant u_i \leqslant Q \quad \forall i \in V_C,
\end{aligned} \tag{8}$$

where $u \in \mathbb{R}^n_+$ represents a vector of auxiliary variables. The first constraint and the first part of the second constraint ensure that for each customer $i \in V_C$, the value of u_i is at least as large as the cumulative customer demands served by the vehicle that visits customer i, immediately after it has left customer i. If all customer demands are strictly positive, then the first constraint ensures that u_j is strictly larger than u_i if $x_{ij} = 1$, $(i,j) \in A$, which excludes subtours. Observation 1 then implies that the combination of the degree constraints (4) and the deterministic MTZ constraints ensures that x induces a set of routes $\mathbf{R}(x)$. Also, the second part of the second deterministic MTZ constraint ensures that the vehicles' capacities are obeyed, that is, $\mathbf{R}(x)$ is feasible in $\mathcal{CVRP}$.

We now derive the *robust MTZ constraints*:

$$\left.\begin{aligned}
& u_j(q) - u_i(q) + Q(1-x_{ij}) \geqslant q_j \quad \forall i, j \in V_C,\ i \neq j \\
& q_i \leqslant u_i(q) \leqslant Q \quad \forall i \in V_C
\end{aligned}\right\} \quad \forall q \in \mathcal{Q}. \tag{9}$$

Here, $u\colon \mathcal{Q} \mapsto \mathbb{R}^n_+$ represents a vector of auxiliary functional variables with the interpretation that $u_i(q)$ denotes the cumulative customer demands served by the vehicle that visits customer $i \in V_C$, immediately after it has left customer i, under the demand realization $q \in \mathcal{Q}$. Note that the robust MTZ constraints are almost identical to their deterministic counterpart, the only differences being the use of auxiliary functions u and the constraint parameterization in $q \in \mathcal{Q}$.

Any robust feasible set of routes $\mathbf{R}(x)$ must obey the vehicles' capacities for any realization $q \in \mathcal{Q}$ of the uncertain demand. Hence, for any robust feasible set of routes $\mathbf{R}(x)$ there are functional variables u that satisfy the robust MTZ constraints (9), which implies that the robust MTZ constraints are *necessary* for robust feasibility. Likewise, due to the conditions (C1) and (C2), the robust MTZ constraints ensure that if $x_{ij} = 1$, $(i,j) \in A$, then $u_j(q) \geqslant u_i(q)$ for all $q \in \mathcal{Q}$ and $u_j(q) \neq u_i(q)$ for some $q \in \mathcal{Q}$. The robust MTZ constraints therefore exclude subtours, and Observation 1 implies that the combination of the degree constraints (4) and the robust MTZ constraints (9) ensures that x induces a set of routes $\mathbf{R}(x)$. The second part of the second constraint in (9) implies that $\mathbf{R}(x)$ also satisfies the vehicles' capacity constraints for all $q \in \mathcal{Q}$, that is, $\mathbf{R}(x)$ is feasible in $\mathcal{RVRP}$. We thus infer that the robust MTZ constraints are also *sufficient* for robust feasibility, and we arrive at the following conclusion.

Observation 2. For any support $\mathcal{Q}$ and binary variables x_{ij}, $(i,j) \in A$, we have that

(i) the robust MTZ constraints (9) are *necessary* to induce a robust feasible set of routes $\mathbf{R}(x)$;

(ii) in conjunction with the degree constraints (4), the robust MTZ constraints (9) are *sufficient* to induce a robust feasible set of routes $\mathbf{R}(x)$.

The robust MTZ constraints in the form (9) require the introduction of n decision variables for each demand realization $q \in \mathcal{Q}$. Because there are infinitely many demand realizations in general, the robust MTZ constraints require the solution of an infinite-dimensional optimization problem. In the following we show that the constraints (9) have an equivalent representation in finitely many decision variables. To this end, we exploit the fact that the constraints (9) are satisfiable by any functions u if and only if they are satisfiable by the linear functions $u'\colon \mathcal{Q} \mapsto \mathbb{R}^n_+$ defined through

$$u'_j(q) = \sum_{i=1}^{l} q_{R_{k,i}(x)} \quad \text{for } j = R_{k,l}(x),\ k \in K,\ 1 \leqslant l \leqslant n_k.$$

Here, $u'_j(q)$ can be interpreted as the cumulative customer demands served by the vehicle that visits customer $j \in V_C$, immediately after it has left customer j, under the demand realization $q \in \mathcal{Q}$. Hence, we can restrict ourselves to the finite-dimensional subspace of linear functions.

Proposition 4. *For any support $\mathcal{Q}$ and binary variables x_{ij}, $(i,j) \in A$, that satisfy the degree constraints* (4), *the*

robust MTZ constraints (9) *are satisfiable by functional decision variables* $u\colon \mathcal{Q} \mapsto \mathbb{R}^n_+$ *if and only if the constraints*

$$\left.\begin{aligned} &\sum_{l\in V_C}(v_{jl}-v_{il})q_l + Q(1-x_{ij}) \geqslant q_j \quad \forall i,\, j\in V_C,\, i\neq j \\ &q_i \leqslant \sum_{l\in V_C} v_{il}q_l \leqslant Q \quad \forall i\in V_C \end{aligned}\right\} \quad \forall q\in\mathcal{Q} \qquad (10)$$

are satisfiable by decision variables $v\in\mathbb{R}^{n\times n}$.

Remark 1. The minimization of the cumulative transportation cost subject to the degree constraints (4) and the robust MTZ constraints (9) can be interpreted as a two-stage robust optimization problem where the routing decision x is taken before the uncertain demands $q\in\mathcal{Q}$ are realized, while the delivery decision u is chosen after the uncertain demands are observed (Ben-Tal et al. 2009, 2004; Chen et al. 2008). In general, the optimal second-stage decision of a two-stage robust optimization problem exhibits a nonlinear dependence on the uncertain parameters (Bertsimas et al. 2010, Kuhn et al. 2011), and linear approximations such as the one adopted in Proposition 4 constitute conservative approximations. Proposition 4 shows that in our setting, a linear approximation is indeed sufficient to describe the optimal second-stage decision.

The minimization of the total transportation cost subject to the degree constraints (4) and the constraints (10) represents a semi-infinite optimization problem as it involves finitely many variables and infinitely many constraints. If the support $\mathcal{Q}$ satisfies condition (C3), then standard robust optimization techniques allow us to equivalently express the problem as a finite-dimensional optimization problem. Indeed, we can reformulate a semi-infinite constraint $q^\top f(\zeta)\leqslant b(\zeta)$ in the customer demands q, decision variables $\zeta\in\mathcal{Z}$ and functions $f\colon \mathcal{Z}\mapsto\mathbb{R}^n$, $b\colon\mathcal{Z}\mapsto\mathbb{R}$ as follows.

$$\begin{aligned} &q^\top f(\zeta)\leqslant b(\zeta) \quad \forall q\in\mathcal{Q} \\ &\iff \max_{q\in\mathbb{R}^n_+}\{q^\top f(\zeta)\colon Wq\leqslant h\}\leqslant b(\zeta) \\ &\iff \min_{\lambda\in\mathbb{R}^r_+}\{h^\top\lambda\colon W^\top\lambda\geqslant f(\zeta)\}\leqslant b(\zeta) \\ &\iff h^\top\lambda\leqslant b(\zeta),\ W^\top\lambda\geqslant f(\zeta) \quad \text{for some } \lambda\in\mathbb{R}^r_+. \end{aligned}$$

The first equivalence follows from the definition of $\mathcal{Q}$ in (C3), while the second equivalence holds due to linear programming duality. Using the same trick, we can express the robust MTZ model as described in Formulation (MTZ).

Formulation (MTZ). Robust Miller-Tucker-Zemlin formulation.

$$\begin{aligned} \underset{x,v,\psi,\omega,\vartheta}{\text{minimize}} \quad & \sum_{(i,j)\in A} c_{ij}x_{ij}, \\ \text{subject to} \quad & x_{ij}\in\{0,1\},\ (i,j)\in A,\ v\in\mathbb{R}^{n\times n}, \\ & \psi_{ijs}\in\mathbb{R}_+,\ i,j\in V_C,\ i\neq j,\ s=1,\ldots,r, \\ & \omega,\vartheta\in\mathbb{R}^{n\times r}_+ \\ & \sum_{\substack{j\in V:\\(i,j)\in A}} x_{ij} = \sum_{\substack{j\in V:\\(j,i)\in A}} x_{ji} = \delta_i \quad \forall i\in V \\ & \sum_{s=1}^r h_s\psi_{ijs}\leqslant Q(1-x_{ij}) \quad \forall i,\,j\in V_C,\ i\neq j \\ & \sum_{s=1}^r W_{sl}\psi_{ijs}\geqslant v_{il}-v_{jl}+\mathbb{1}_{[l=j]} \\ & \qquad\qquad \forall i,\,j,\,l\in V_C,\ i\neq j \\ & \sum_{s=1}^r h_s\omega_{is}\leqslant 0 \quad \forall i\in V_C \\ & \sum_{s=1}^r W_{sl}\omega_{is}\geqslant \mathbb{1}_{[l=i]}-v_{il} \quad \forall i,\,l\in V_C \\ & \sum_{s=1}^r h_s\vartheta_{is}\leqslant Q \quad \forall i\in V_C \\ & \sum_{s=1}^r W_{sl}\vartheta_{is}\geqslant v_{il} \quad \forall i,\,l\in V_C. \end{aligned}$$

Remark 2. A tighter version of Formulation (MTZ) can be obtained if we derive the robust counterpart of the so-called *lifted* deterministic Miller-Tucker-Zemlin constraints (Desrochers and Laporte 1991, Kara et al. 2004). For brevity, we omit the largely similar derivation.

4.2.2. Robust Precedence Formulation. The reformulated robust MTZ constraints (10) from Proposition 4 require the dualization of n^2+n constraints. For supports $\mathcal{Q}$ that satisfy condition (C3), this results in n^2r+nr dual variables and n^3+2n^2+n potentially dense constraints. In the following proposition we exploit our structural knowledge about the optimal solution to the constraints (10) to obtain a sparser formulation.

Proposition 5. *For any support* $\mathcal{Q}$ *and binary variables* x_{ij}, $(i,j)\in A$ *that satisfy the degree constraints* (4), *the constraints* (10) *are satisfiable by* $v\in\mathbb{R}^{n\times n}$ *if and only if the constraints*

$$\begin{aligned} &x_{ij}\leqslant p_{ij}\leqslant\min\{1-x_{ji},1-x_{0j}\} \quad \forall i,\,j\in V_C,\ i\neq j \\ &1-x_{ij}\geqslant\max\{p_{li}-p_{lj},p_{lj}-p_{li}\} \quad \forall i,\,j\in V_C,\ i\neq j, \\ &\qquad\qquad \forall l\in V_C\backslash\{i,j\}, \qquad (11) \\ &\sum_{j\in V_C} p_{ji}q_j\leqslant Q \quad \forall i\in V_C,\ q\in\mathcal{Q} \end{aligned}$$

are satisfiable by $p\in\{0,1\}^{n\times n}$, *where* $p_{ii}=1$ *for* $i\in V_C$.

The constraints (11) require the dualization of only n constraints and result in nr dual variables and $2n^3-2n^2+2n$ constraints, out of which at most n^2+n constraints are dense. It follows from Observation 2 and Proposition 4 that for any support $\mathcal{Q}$ and binary variables x_{ij}, $(i,j)\in A$, the combination of constraint set (11) and the degree constraints (4) is necessary and sufficient for the robust feasibility of $\mathbf{R}(x)$. Interestingly, the new decision variables p_{ij} allow for a natural interpretation as precedence variables.

PROPOSITION 6. *For any support $\mathcal{Q}$ and any solution (x, p), $x_{ij} \in \{0,1\}$, $(i,j) \in A$, $p \in \{0,1\}^{n\times n}$, $p_{ii} = 1$ for $i \in V_C$, that satisfies the degree constraints* (4) *and the constraints* (11), *we have $p_{ij} = 1$ if and only if customers i and j are served by the same vehicle and customer i is served first.*

Although the reformulation from Proposition 5 requires fewer dual variables and fewer potentially dense constraints, it requires the introduction of $n^2 - n$ binary variables p_{ij}, $i \neq j$. However, the proof of Proposition 6 allows us to weaken the assumption that p is binary, and we arrive at the following conclusion.

COROLLARY 1. *For any support $\mathcal{Q}$ and binary variables $x_{ij} \in \{0,1\}$, $(i,j) \in A$, that satisfy the degree constraints* (4), *the constraints* (11) *are satisfiable by binary variables $p \in \{0,1\}^{n\times n}$, $p_{ii} = 1$ for $i \in V_C$, if and only if they are satisfiable by continuous variables $p \in [0,1]^{n\times n}$, $p_{ii} = 1$ for $i \in V_C$.*

We remark that we can immediately derive a deterministic version of the robust precedence constraints (11) if we disregard the parameterization of the last constraint in the customer demands $q \in \mathcal{Q}$. To our knowledge, this would result in a new formulation for the deterministic CVRP, which might be of interest in its own right.

For supports $\mathcal{Q}$ that satisfy condition (C3), Corollary 1 allows us to express the robust precedence formulation as the finite-dimensional model described in Formulation (PRE). We close this section with a discussion of some inequalities that can be used to further tighten this formulation.

Formulation (PRE). Robust precedence formulation.

$$\begin{aligned}
\underset{x,p,\psi}{\text{minimize}} \quad & \sum_{(i,j)\in A} c_{ij}x_{ij}, \\
\text{subject to} \quad & x_{ij} \in \{0,1\},\ (i,j)\in A,\ p \in [0,1]^{n\times n},\ \psi \in \mathbb{R}_+^{n\times r} \\
& \sum_{\substack{j\in V:\\ (i,j)\in A}} x_{ij} = \sum_{\substack{j\in V:\\ (j,i)\in A}} x_{ji} = \delta_i \quad \forall i \in V \\
& x_{ij} \leqslant p_{ij} \leqslant \min\{1 - x_{ji}, 1 - x_{0j}\} \\
& \qquad\qquad \forall i, j \in V_C,\ i \neq j \\
& 1 - x_{ij} \geqslant \max\{p_{li} - p_{lj}, p_{lj} - p_{li}\} \\
& \qquad \forall i, j \in V_C,\ i \neq j,\ l \in V_C\backslash\{i,j\} \\
& \sum_{s=1}^{r} h_s \psi_{is} \leqslant Q \quad \forall i \in V_C \\
& \sum_{s=1}^{r} W_{sl}\psi_{is} \geqslant p_{li} \quad \forall i, l \in V_C \\
& p_{ii} = 1 \quad \forall i \in V_C.
\end{aligned}$$

REMARK 3. Proposition 5 remains valid if we replace the constraint set (11) with

$$x_{ij} \leqslant p_{ij} \leqslant \min\{1 - x_{ji}, 1 - x_{0j}, 1 - x_{i0}\} \quad \forall i, j \in V_C,\ i \neq j$$

$$1 - x_{ij} - x_{ji} \geqslant \max\{p_{li} - p_{lj}, p_{lj} - p_{li}, p_{il} - p_{jl}, p_{jl} - p_{il}\} \quad \forall i, j \in V_C,\ i < j,\ l \in V_C\backslash\{i,j\}, \tag{11'}$$

$$q_i + \sum_{\substack{j\in V_C,\\ j\neq i}} (p_{ij} + p_{ji})q_j \leqslant Q \quad \forall i \in V_C,\ q \in \mathcal{Q},$$

and also add the constraint

$$p_{ij} + p_{ji} \leqslant 1 \quad \forall i, j \in V_C,\ i < j.$$

While the set of binary variables x_{ij}, $(i,j) \in A$, that satisfy constraints (11) does not change if we replace (11) with (11′), the new formulation can help improve the bounds obtained from continuous relaxations $x_{ij} \in [0,1]$ of the robust precedence formulation.

4.3. Commodity Flow Formulations

Both the deterministic and the robust one-commodity flow (1CF) constraints encode the cumulative customer demands served by the vehicle that traverses the arc $(i,j) \in A$, immediately after it has left the (customer or depot) node $i \in V$. The *deterministic 1CF constraints* (Gouveia 1995) achieve this through auxiliary variables $f_{ij} \in \mathbb{R}_+$, $(i,j) \in A$, and the constraints

$$\begin{aligned}
& \sum_{\substack{i\in V:\\ (j,i)\in A}} f_{ji} = \sum_{\substack{i\in V:\\ (i,j)\in A}} f_{ij} + q_j \quad \forall j \in V_C, \\
& q_i x_{ij} \leqslant f_{ij} \leqslant (Q - q_j)x_{ij} \quad \forall (i,j) \in A,
\end{aligned} \tag{12}$$

where we set $q_0 = 0$. These constraints can be tightened by setting $f_{0j} = 0$ for all $j \in V_C$. Assume that the values of x_{ij} form a subtour, that is, $x_{i_1i_2} = x_{i_2i_3} = \cdots = x_{i_{t-1}i_t} = x_{i_ti_1} = 1$ for some $\{i_1, \ldots, i_t\} \subseteq V_C$. Together with the degree constraints (4), the second constraint in (12) then implies that $f_{i_1i_2}, f_{i_2i_3}, \ldots, f_{i_{t-1}i_t}, f_{i_ti_1}$ are the only variables in the set $\{f_{ij}\colon \{i,j\} \cap \{i_1, \ldots, i_t\} \neq \varnothing\}$ that can be nonzero. Hence, if all customer demands are strictly positive, then the first constraint in (12) requires that $f_{i_ti_1} > f_{i_{t-1}i_t} > \cdots > f_{i_1i_2} > f_{i_ti_1}$, which is a contradiction. The deterministic 1CF constraints thus exclude subtours, and the combination of the degree constraints (4) and the deterministic 1CF constraints (12) ensures that x induces a set of routes $\mathbf{R}(x)$ by virtue of Observation 1. Moreover, the first constraint and the first part of the second constraint in (12) ensure that for each arc $(i,j) \in A$, the value of f_{ij} is at least as large as the cumulative customer demands served by the vehicle that traverses the arc (i,j), immediately after it has left the (customer or depot) node $i \in V$. Thus, the second part of the second constraint in (12) implies that the capacities of all vehicles are obeyed, that is, $\mathbf{R}(x)$ is feasible in $\mathcal{CVRP}$.

We now derive the *robust 1CF constraints*:

$$\left.\begin{aligned}
& \sum_{\substack{i\in V:\\ (j,i)\in A}} f_{ji}(q) = \sum_{\substack{i\in V:\\ (i,j)\in A}} f_{ij}(q) + q_j \quad \forall j \in V_C \\
& q_i x_{ij} \leqslant f_{ij}(q) \leqslant (Q - q_j)x_{ij} \quad \forall (i,j) \in A
\end{aligned}\right\} \quad \forall q \in \mathcal{Q}. \tag{13}$$

Here, $f_{ij}\colon \mathcal{Q} \mapsto \mathbb{R}_+$, $(i,j) \in A$, represent auxiliary functional variables with the interpretation that $f_{ij}(q)$ denotes

the cumulative customer demands served by the vehicle that traverses the arc $(i, j) \in A$, immediately after it has left the (customer or depot) node $i \in V$, under the demand realization $q \in \mathcal{Q}$. Similar to the robust MTZ constraints, the robust 1CF constraints are almost identical to their deterministic counterpart, the only differences being the use of functional auxiliary variables and the constraint parameterization in the uncertain customer demands $q \in \mathcal{Q}$.

Any robust feasible set of routes $\mathbf{R}(x)$ must satisfy the capacity constraints for any realization $q \in \mathcal{Q}$ of the uncertain demand. Hence, for any robust feasible set of routes $\mathbf{R}(x)$ there are functional variables f_{ij}, $(i, j) \in A$, that satisfy the robust 1CF constraints (13). The robust 1CF constraints are therefore *necessary* for robust feasibility. Likewise, due to (C1) and (C2), the robust 1CF constraints imply that for each arc $(i, j) \in A$, $\sum_i f_{ji}(q) \geqslant \sum_i f_{ij}(q)$ for all $q \in \mathcal{Q}$ and $\sum_i f_{ji}(q) \neq \sum_i f_{ij}(q)$ for some $q \in \mathcal{Q}$. A similar argument as for the deterministic 1CF constraints therefore shows that the combination of the degree constraints (4) and the robust 1CF constraints (13) ensures that x induces a set of routes $\mathbf{R}(x)$. Moreover, the robust 1CF constraints imply that for each arc $(i, j) \in A$, the value of $f_{ij}(q)$ is at least as large as the cumulative customer demands served by the vehicle that traverses the arc $(i, j) \in A$, immediately after it has left the (customer or depot) node $i \in V$, under demand realization $q \in \mathcal{Q}$. Thus, the second part of the second constraint in (13) ensures that the capacity constraints of all vehicles are satisfied for all customer demands $q \in \mathcal{Q}$, that is, $\mathbf{R}(x)$ is feasible in $\mathcal{RVRP}$. We thus infer that the robust 1CF constraints are also *sufficient* for robust feasibility, and we arrive at the following conclusion.

Observation 3. For any support $\mathcal{Q}$ and binary variables x_{ij}, $(i, j) \in A$, we have that

(i) the robust 1CF constraints (13) are *necessary* to induce a robust feasible set of routes $\mathbf{R}(x)$;

(ii) in conjunction with the degree constraints (4), the robust 1CF constraints (13) are *sufficient* to induce a robust feasible set of routes $\mathbf{R}(x)$.

The robust 1CF constraints (13) require $n^2 + n$ decision variables $f_{ij}(q)$ for each demand realization $q \in \mathcal{Q}$. Because there are infinitely many demand realizations in general, the robust 1CF constraints require the solution of an infinite-dimensional optimization problem. In the following we show that the constraints (13) have an equivalent representation in finitely many decision variables. To this end, we exploit the fact that the constraints (13) are satisfiable by any functions f_{ij} if and only if they are satisfiable by the linear functions f'_{ij}: $\mathcal{Q} \mapsto \mathbb{R}_+$, $(i, j) \in A$, defined through

$$f'_{ij}(q) = x_{ij} \sum_{s=1}^{l} q_{R_{k,s}(x)} \quad \text{for } i = R_{k,l}(x),\ k \in K,\ 1 \leqslant l \leqslant n_k.$$

Here, $f'_{ij}(q)$ can be interpreted as the cumulative customer demands served by the vehicle that traverses the arc $(i, j) \in A$, immediately after it has left the (customer or depot) node $i \in V$, under the demand realization $q \in \mathcal{Q}$. Hence, we can restrict ourselves to the finite-dimensional subspace of linear functions.

Proposition 7. *For any support $\mathcal{Q}$ and binary variables x_{ij}, $(i, j) \in A$, that satisfy the degree constraints* (4), *the robust 1CF constraints* (13) *are satisfiable by functional decision variables f_{ij}: $\mathcal{Q} \mapsto \mathbb{R}_+$, $(i, j) \in A$, if and only if the constraints*

$$\begin{aligned} &\sum_{\substack{i \in V:\\ i \neq j}} (g_{jil} - g_{ijl}) = \mathbb{1}_{[l=j]} \quad \forall j, l \in V_C, \\ &q_i x_{ij} \leqslant \sum_{l \in V_C} g_{ijl} q_l \leqslant (Q - q_j) x_{ij} \quad \forall (i, j) \in A,\ \forall q \in \mathcal{Q} \end{aligned} \tag{14}$$

are satisfiable by decision variables $g_{ijl} \in \mathbb{R}$, $(i, j) \in A$ and $l \in V_C$.

Formulation (1CF). Robust one-commodity flow formulation.

$$\begin{aligned} \underset{x, g, \psi, \omega}{\text{minimize}} \quad & \sum_{(i,j) \in A} c_{ij} x_{ij}, \\ \text{subject to} \quad & x_{ij} \in \{0, 1\},\ (i, j) \in A,\ g_{ijl} \in \mathbb{R},\ (i, j) \in A,\ l \in V_C, \\ & \psi_{ijs}, \omega_{ijs} \in \mathbb{R}_+,\ (i, j) \in A,\ s = 1, \ldots, r, \\ & \sum_{\substack{j \in V:\\ (i,j) \in A}} x_{ij} = \sum_{\substack{j \in V:\\ (j,i) \in A}} x_{ji} = \delta_i \quad \forall i \in V, \\ & \sum_{\substack{i \in V:\\ i \neq j}} (g_{jil} - g_{ijl}) = \mathbb{1}_{[l=j]} \quad \forall j, l \in V_C, \\ & \sum_{s=1}^{r} h_s \psi_{ijs} \leqslant 0 \quad \forall (i, j) \in A, \\ & \sum_{s=1}^{r} W_{sl} \psi_{ijs} \geqslant \mathbb{1}_{[l=i]} x_{ij} - g_{ijl} \quad \forall (i, j) \in A,\ \forall l \in V_C, \\ & \sum_{s=1}^{r} h_s \omega_{ijs} \leqslant Q x_{ij} \quad \forall (i, j) \in A, \\ & \sum_{s=1}^{r} W_{sl} \omega_{ijs} \geqslant \mathbb{1}_{[l=j]} x_{ij} + g_{ijl} \quad \forall (i, j) \in A,\ \forall l \in V_C. \end{aligned}$$

Remark 4. Similar to the robust MTZ formulation in §4.2, the minimization of the cumulative transportation cost subject to the degree constraints (4) and the robust 1CF constraints (13) can be interpreted as a two-stage robust optimization problem where the routing decision x is taken before the uncertain demands $q \in \mathcal{Q}$ are realized, while the flow decision f is chosen after the uncertain demands are observed (Ben-Tal et al. 2009, 2004; Chen et al. 2008). Proposition 7 shows that in our setting, there is always an optimal second-stage decision that exhibits a linear dependence on the uncertain demands.

The minimization of the total transportation cost subject to the degree constraints (4) and the constraints (14) constitutes a semi-infinite optimization problem. Under condition (C3), we can reformulate this problem as the finite-dimensional model described in Formulation (1CF).

We close this section with an extension to $\mathcal{RVRP}$ instances with symmetric transportation costs.

REMARK 5. If a $\mathcal{CVRP}$ instance has symmetric transportation costs, that is, if $c_{ij} = c_{ji}$ for all $(i, j) \in A$, then the number of binary variables x_{ij} can be reduced. To this end, we define the extended network $V' = V \cup \{n+1\}$ and $A' = \{(i, j) \in V' \times V' \colon i \neq j\} \setminus \{(0, n+1), (n+1, 0)\}$ that contains a duplicate depot node $n+1$ for the end of each vehicle route. The reduced variable set x_{ij}, $(i, j) \in A'$ and $i < j$, induces a feasible solution to $\mathcal{CVRP}$ if and only if it satisfies the symmetric degree constraints

$$\begin{aligned} &\sum_{\substack{j\in V':\\ j<i}} x_{ji} + \sum_{\substack{j\in V':\\ j>i}} x_{ij} = 2 \quad \forall i \in V_C, \\ &\sum_{j\in V_C} x_{0j} = \sum_{j\in V_C} x_{j(n+1)} = m, \end{aligned} \tag{4'}$$

and the *deterministic two-commodity flow (2CF) constraints* (Baldacci et al. 2004):

$$\begin{aligned} &f_{ij} + f_{ji} = Qx_{ij} \quad \forall (i, j) \in A',\ i < j, \\ &\sum_{\substack{j\in V':\\ i\neq j}} f_{ij} = Q - q_i \quad \forall i \in V_C, \\ &\sum_{j\in V_C} f_{0j} = \sum_{i\in V_C} q_i, \end{aligned}$$

where $f_{ij} \in \mathbb{R}_+$ for all $(i, j) \in A'$. The constraints can be tightened by setting $f_{i(n+1)} = 0$ for all $i \in V_C$. Similar to the 1CF constraints, we can derive the *robust 2CF constraints*:

$$\begin{aligned} &g^0_{ij} + g^0_{ji} = Qx_{ij}, \quad g_{ijl} + g_{jil} = 0 \quad \forall (i, j) \in A',\ i < j,\ \forall l \in V_C, \\ &\sum_{\substack{j\in V':\\ i\neq j}} g^0_{ij} = Q, \quad \sum_{\substack{j\in V':\\ i\neq j}} g_{ijl} + \mathbb{1}_{[l=i]} = 0 \quad \forall i,\ l \in V_C, \\ &\sum_{j\in V_C} g^0_{0j} = 0, \quad \sum_{j\in V_C} g_{0jl} = 1 \quad \forall l \in V_C, \\ &g^0_{ij} + \sum_{l\in V_C} g_{ijl} q_l \geqslant 0 \quad \forall (i, j) \in A',\ \forall q \in \mathcal{Q}, \end{aligned}$$

where g^0_{ij}, $g_{ijl} \in \mathbb{R}$ for all $(i, j) \in A'$ and $l \in V_C$. In analogy to the 1CF constraints, one can show that x induces a feasible solution to $\mathcal{RVRP}$ if and only if it satisfies the symmetric degree constraints (4′) and the robust 2CF constraints. For supports $\mathcal{Q}$ that satisfy condition (C3), the 2CF problem can be reformulated as the finite-dimensional model described in Formulation (2CF).

Formulation (2CF). Two-commodity flow formulation.

$$\begin{aligned} \underset{x, g^0, g, \psi}{\text{minimize}} \quad & \sum_{(i,j)\in A'} c_{ij} x_{ij}, \\ \text{subject to} \quad & x_{ij} \in \{0, 1\},\ (i, j) \in A',\ i < j,\ g^0_{ij}, \\ & g_{ijl} \in \mathbb{R},\ (i, j) \in A',\ l \in V_C, \\ & \psi_{ijs} \in \mathbb{R}_+,\ (i, j) \in A',\ s = 1, \ldots, r, \\ & \sum_{\substack{j\in V':\\ j<i}} x_{ji} + \sum_{\substack{j\in V':\\ j>i}} x_{ij} = 2 \quad \forall i \in V_C, \\ & \sum_{j\in V_C} x_{0j} = \sum_{j\in V_C} x_{j(n+1)} = m, \\ & g^0_{ij} + g^0_{ji} = Qx_{ij}, \quad g_{ijl} + g_{jil} = 0 \\ & \qquad\qquad \forall (i, j) \in A',\ i < j,\ \forall l \in V_C, \\ & \sum_{\substack{j\in V':\\ i\neq j}} g^0_{ij} = Q, \quad \sum_{\substack{j\in V':\\ i\neq j}} g_{ijl} + \mathbb{1}_{[l=i]} = 0 \quad \forall i, l \in V_C, \\ & \sum_{j\in V_C} g^0_{0j} = 0, \quad \sum_{j\in V_C} g_{0jl} = 1 \quad \forall l \in V_C, \\ & \sum_{s=1}^{r} h_s \psi_{ijs} \leqslant g^0_{ij} \quad \forall (i, j) \in A', \\ & \sum_{s=1}^{r} W_{sl} \psi_{ijs} \geqslant -g_{ijl} \quad \forall (i, j) \in A',\ \forall l \in V_C. \end{aligned}$$

4.4. Vehicle Assignment Formulations

In this subsection, we develop a new class of $\mathcal{CVRP}$ formulations that is motivated by the three-index vehicle flow formulation (Golden et al. 1977) and that contains the customer demands only in a small number of constraints. It turns out that the resulting $\mathcal{RVRP}$ formulations contain fewer dense constraints and dual variables than the formulations from the previous subsections. They are also particularly useful in solving the chance-constrained CVRP presented in §EC.3 of the paper's e-companion. While in this paper we restrict our attention to robust vehicle assignment formulations, we remark that the vehicle assignment formulations might also be of interest in deterministic CVRP variants where the explicit assignment of customers to vehicles is required, for example the consistent vehicle routing problem (Groër et al. 2009, Tarantilis et al. 2012).

The new formulations are based on the introduction of auxiliary binary variables z_{ik} that indicate whether customer $i \in V_C$ is served by vehicle $k \in K$. For a given routing decision x, we thus intend to set $z_{ik} = \mathbb{1}_{[i\in R_k(x)]}$. The following proposition presents a constraint set that enforces this definition.

PROPOSITION 8. *For any binary variables x_{ij}, $(i, j) \in A$, that satisfy the degree constraints* (4) *and the subtour*

constraints (5), $z \in \{0,1\}^{n\times m}$ *satisfies the vehicle assignment* (*VA*) *constraints*

$$\begin{aligned} &1 - x_{ij} - x_{ji} \geqslant z_{ik} - z_{jk} \quad \forall i, j \in V_C,\ i \neq j,\ \forall k \in K,\\ &z_{ik} + \sum_{l\in\{1,\ldots,k\}} z_{jl} \leqslant 3 - x_{0i} - x_{0j}\\ &\qquad\qquad \forall i, j \in V_C,\ i < j,\ \forall k \in K,\\ &\sum_{k\in K} z_{ik} = 1 \quad \forall i \in V_C, \end{aligned} \tag{15}$$

if and only if $z_{ik} = \mathbb{1}_{[i\in R_k(x)]}$, *where* $\mathbf{R}(x)$ *satisfies* $R_{1,1}(x) < R_{2,1}(x) < \cdots < R_{m,1}(x)$.

The VA constraints (15) ensure that the routes encoded by z_{ik} are ordered lexicographically with respect to their first customers. This is desirable as it eliminates redundant symmetries in the feasible region for z. Note that the constraints require the introduction of nm binary variables z_{ik}, $i \in V_C$ and $k \in K$. The following proposition shows that we can relax this binarity requirement.

Proposition 9. *Assume that the binary variables* x_{ij}, $(i,j) \in A$, *satisfy the degree constraints* (4) *and the subtour constraints* (5). *Then the VA constraints* (15) *are satisfiable by a variable vector* $z \in \{0,1\}^{n\times m}$ *if and only if they are satisfiable by a variable vector* $z \in [0,1]^{n\times m}$.

By virtue of Observation 1, the binary vector x induces a set of routes $\mathbf{R}(x)$ if and only if it satisfies the degree constraints (4) and the subtour constraints (5). In that case, the variable vector $z \in [0,1]^{n\times m}$ satisfies $z_{ik} = \mathbb{1}_{[i\in R_k(x)]}$ if and only if (x,z) satisfies the VA constraints (15). Hence, if (x,z) satisfies the degree constraints (4), the subtour constraints (5) and the VA constraints (15), then the set of routes $\mathbf{R}(x)$ is feasible if and only if z satisfies the vehicle capacity constraints

$$\sum_{i\in V_C} q_i z_{ik} \leqslant Q \quad \forall k \in K.$$

Likewise, if (x,z) satisfies the degree constraints (4), the subtour constraints (5) and the VA constraints (15), then the set of routes $\mathbf{R}(x)$ is robust feasible if and only if z satisfies the semi-infinite constraints

$$\sum_{i\in V_C} q_i z_{ik} \leqslant Q \quad \forall k \in K,\ \forall q \in \mathcal{Q}. \tag{16}$$

We summarize this conclusion as follows.

Observation 4. For any support $\mathcal{Q}$, binary variables x_{ij}, $(i,j) \in A$, and $z \in [0,1]^{n\times m}$, we have

(i) the VA constraints (15) and the vehicle capacity constraints (16) are *necessary* to induce a robust feasible set of routes $\mathbf{R}(x)$;

(ii) in conjunction with the degree constraints (4) and the subtour constraints (5), the VA constraints (15) and the vehicle capacity constraints (16) are *sufficient* to induce a robust feasible set of routes $\mathbf{R}(x)$.

For supports $\mathcal{Q}$ that satisfy condition (C3), we can then use robust optimization techniques to obtain the vehicle assignment model described in Formulation (VA). Although Formulation (VA) contains the degree constraints (4) and the VA constraints (15), it does not exclude subtours. We therefore amend this formulation with the robust RCI constraints (7) or the deterministic MTZ constraints (8). We refer to these two variants as Formulations (VA1) and (VA2), respectively.

Formulation (VA). Robust vehicle assignment formulation.

$$\begin{aligned} \underset{x,z,\psi}{\text{minimize}} \quad & \sum_{(i,j)\in A} c_{ij} x_{ij},\\ \text{subject to} \quad & x_{ij} \in \{0,1\},\ (i,j) \in A,\ z \in [0,1]^{n\times m},\ \psi \in \mathbb{R}_+^{m\times r},\\ & \sum_{\substack{j\in V:\\ (i,j)\in A}} x_{ij} = \sum_{\substack{j\in V:\\ (j,i)\in A}} x_{ji} = \delta_i \quad \forall i \in V\\ & 1 - x_{ij} - x_{ji} \geqslant z_{ik} - z_{jk}\\ & \qquad \forall i, j \in V_C,\ i \neq j,\ \forall k \in K,\\ & z_{ik} + \sum_{l\in\{1,\ldots,k\}} z_{jl} \leqslant 3 - x_{0i} - x_{0j}\\ & \qquad \forall i, j \in V_C,\ i < j,\ \forall k \in K,\\ & \sum_{k\in K} z_{ik} = 1 \quad \forall i \in V_C,\\ & \sum_{s=1}^{r} h_s \psi_{ks} \leqslant Q \quad \forall k \in K,\\ & \sum_{s=1}^{r} W_{sl} \psi_{ks} \geqslant z_{lk} \quad \forall l \in V_C,\ \forall k \in K. \end{aligned}$$

5. Robust Rounded Capacity Inequality (RCI) Cuts

To date, branch-and-cut and branch-and-price are the most efficient exact solution methods for $\mathcal{CVRP}$. In §6, we use a branch-and-cut algorithm to solve the proposed $\mathcal{RVRP}$ formulations. Branch-and-cut methods combine the techniques of branch-and-bound and cutting planes. A branch-and-bound algorithm solves $\mathcal{RVRP}$ by implicitly enumerating the feasible solutions in a tree. Branch-and-cut methods extend this principle by applying a cutting plane algorithm to each tree node to add constraints (so-called *cuts*) that are satisfied by all feasible solutions to $\mathcal{RVRP}$ but that are violated by the node solution. These cuts help exclude infeasible solutions and thereby expedite the search process.

It is well known that the deterministic RCI cuts (6) can speed up the solution of $\mathcal{CVRP}$; see Toth and Vigo (2002). In §6, we use the robust RCI constraints (7) as cuts in a branch-and-cut algorithm for $\mathcal{RVRP}$. The identification of a generic customer set $S \subseteq V_C$ that violates its associated RCI constraint for a fractional routing decision x is nontrivial. For $\mathcal{CVRP}$, this typically is achieved by a

local search that aims to find a set S whose RCI constraint is maximally violated. For example, Lysgaard et al. (2004) and Ralphs et al. (2003) present heuristics that consider the graph $G(x) = (V_C, \{(i,j) \in A\colon x_{ij} > 0\})$ induced by the node solution x. The algorithms form sets of customers that participate in components of this graph and check the corresponding RCI constraints for violations. Augerat et al. (1998) and Gounaris et al. (2011) present tabu search heuristics that iteratively construct customer sets through a sequence of greedy perturbations (e.g., by adding or removing a single customer). In §6, we use a variant of the algorithm presented in Gounaris et al. (2011) to identify violated robust RCI cuts in our branch-and-cut scheme. Our algorithm starts with a randomly selected customer set S and iteratively perturbs this set through a sequence of perturbations in which individual customers are added or removed. In each iteration, the customer is selected in a greedy fashion, that is, we choose a customer whose inclusion or removal maximizes the difference between the right-hand side and the left-hand side of the corresponding robust RCI constraint (7). We also maintain Tabu lists of customers that have recently been added or removed to avoid cycling of the procedure. We terminate the procedure when we are unable to increase the difference between the right-hand side and the left-hand side of constraint (7) for a number of consecutive iterations.

By construction, Formulations (2VF) and (VA1) already contain all robust RCI constraints. As a result, both formulations are very large and impractical to solve. In our numerical tests, we therefore remove the robust RCI constraints from (2VF) and (VA1) and dynamically reintroduce them with our separation heuristic. Because the heuristic might miss a violated constraint, the branch-and-cut algorithm must check the feasibility of each binary node solution in (2VF) and (VA1). If a node solution contains a customer set $S \subseteq V_C$ that forms a subtour, then we include the robust RCI cut corresponding to S and resolve the node problem. We do the same when a node solution to (2VF) contains a customer set $S \subseteq V_C$ that forms a capacity-infeasible route.

During the solution of $\mathcal{RVRP}$, our branch-and-cut algorithm needs to frequently check for violations of the robust RCI constraints (7). Hence, it is crucial that we can efficiently evaluate the right-hand sides of these constraints. In general, this requires the solution of a linear program. In the remainder of this section, we provide analytical solutions to these linear programs for two broad classes of demand supports $\mathcal{Q}$, which we call *budget supports* and *factor model supports*.

5.1. Budget Supports

Consider first the case where the uncertain customer demands are supported on the set

$$\mathcal{Q} = \left\{ q \in \mathbb{R}^n_+ \colon q \in [\underline{q}, \bar{q}],\ \sum_{i \in B_l} q_i \leqslant b_l \text{ for } l = 1, \ldots, L \right\}, \tag{17}$$

which results from the intersection of the n-dimensional hyper-rectangle $[\underline{q}, \bar{q}]$ with L budget constraints involving customer subsets $B_l \subseteq V_C$. Here, $\underline{q}, \bar{q} \in \mathbb{R}^n_+$, $L \in \mathbb{N}$, $B_l \subseteq V_C$ and $b_l \in \mathbb{R}_+$ are parameters that need to be specified by the decision maker.

To evaluate the right-hand side of the robust RCI constraints (7) for supports of type (17), we need to maximize $\sum_{i \in S} q_i$ over the polytope defined in (17) for some customer subset $S \subseteq V_C$. The following proposition shows that this problem has an analytical solution if the customer subsets B_l participating in the budget constraints in (17) are disjoint.

Proposition 10. *Assume that the sets $\{B_l\}_{l=1}^L$ in (17) are disjoint, that is, $B_l \cap B_{l'} = \varnothing$ for $l \neq l'$. Then, for any customer subset $S \subseteq V_C$, the maximum of $\sum_{i \in S} q_i$ over $\mathcal{Q}$ from (17) is given by*

$$\sum_{i \in S} \underline{q}_i + \sum_{l=1}^{L} \min\left\{ b_l - \sum_{i \in B_l} \underline{q}_i,\ \sum_{i \in S \cap B_l} (\bar{q}_i - \underline{q}_i) \right\} + \sum_{i \in S \setminus \bigcup_{l=1}^{L} B_l} (\bar{q}_i - \underline{q}_i).$$

With suitably chosen data structures, Proposition 10 allows us to maximize $\sum_{i \in S} q_i$ over the polytope (17) in time $\mathcal{O}(|S|)$. Moreover, for two customer subsets S_1, $S_2 \subseteq V_C$, the maximum of $\sum_{i \in S_2} q_i$ over (17) can be calculated from the maximum of $\sum_{i \in S_1} q_i$ over (17) in time $\mathcal{O}(|S_1 \setminus S_2| + |S_2 \setminus S_1|)$. This allows us to evaluate the corresponding robust RCI constraints efficiently by the aforementioned incremental procedures.

5.2. Factor Model Supports

Assume now that the uncertain customer demands are supported on the set

$$\mathcal{Q} = \{ q \in \mathbb{R}^n_+ \colon q = q^0 + \Phi\xi \text{ for some } \xi \in \Xi \}, \tag{18a}$$

where

$$\Xi = \{ \xi \in \mathbb{R}^F \colon \xi \in [-\mathrm{e}, +\mathrm{e}],\ \mathrm{e}^\top \xi \in [-\beta F, +\beta F] \}. \tag{18b}$$

Here, $q^0 \in \mathbb{R}^n_+$, $\Phi \in \mathbb{R}^{n \times F}$, $F \in \mathbb{N}$ and $\beta \in [0, 1]$ are parameters that need to be specified by the decision maker. The support (18) can be interpreted as follows. The demand vector q results from the nominal demand vector q^0 and an additive disturbance $\Phi\xi$ that depends on F independent factors $\xi_1, \ldots, \xi_F$. If the parameter β is set to zero, then the second constraint in (18b) reflects the expectation that as many factors ξ_f will be above the point estimate $0 \in \mathbb{R}^F$ as there will be below the estimate. Supports of the type (18) have recently attracted interest in the context of financial portfolio optimization (Ceria and Stubbs 2006).

The following proposition shows that the maximum of $\sum_{i \in S} q_i$ over the polytope defined in (18) can be determined efficiently for any customer subset $S \subseteq V_C$.

PROPOSITION 11. *For a customer subset $S \subseteq V_C$, assume that $f_1, \ldots, f_F$ represents an ordering of the factors ξ_f in* (18b) *according to nonincreasing marginal demands*

$$\sum_{i\in S} \Phi_{if_1} \geqslant \sum_{i\in S} \Phi_{if_2} \geqslant \cdots \geqslant \sum_{i\in S} \Phi_{if_F}.$$

Then, the maximum of $\sum_{i\in S} q_i$ over the polytope defined in (18) *is given by*

$$\sum_{i\in S} q_i^0 + \min\left\{\sum_{f=1}^{F}\left|\sum_{i\in S}\Phi_{if} - \lambda\right| + \beta F|\lambda|\colon \lambda \in \left\{0, \sum_{i\in S}\Phi_{if_{\ell^+}}, \sum_{i\in S}\Phi_{if_{\ell^-}}\right\}\right\}, \quad (19)$$

where $\ell^+ = \lceil (1+\beta)F/2 \rceil$ and $\ell^- = \max\{\lceil (1-\beta)F/2 \rceil, 1\}$.

With suitably chosen data structures, Proposition 11 allows us to maximize $\sum_{i\in S} q_i$ over the polytope (18) in time $\mathcal{O}(F|S| + F\log F)$. Note that $F \ll n$ for typical factor models, which means that we can evaluate the corresponding robust RCI constraints efficiently.

6. Computational Results

In this computational study, we consider $\mathcal{RVRP}$ instances in which the customer demands are supported on budget and factor model sets; see §5. All instances are based on the standard CVRP benchmark problems presented in Díaz (2011). Following the convention in the literature, we name the problems "*X-nY-kZ*," where X denotes the literature source of the instance, Y the number of nodes in the network (i.e., the number of customers plus one) and Z the number of vehicles. To construct meaningful supports for the customer demands, we use only problems for which two-dimensional coordinates for the customers and the depot are available. The transportation costs c_{ij} result from the Euclidean distance between $i, j \in V$, rounded to the nearest integer, per standard literature practice. We solve the instances with CPLEX 12.1 (IBM Corp 2009) on a single-core Intel 2.66-GHz processor with 3-GB memory. We use the CPLEX default options with two exceptions: (i) we disable the CPLEX-generated generic MIP cuts, and (ii) we introduce the robust RCI cuts (7) globally through user-defined callbacks; see §5.

We partition the customers V_C into four geographic quadrants—NE, NW, SW, SE—based on the coordinates in the benchmark problems. We define the following budget support for the demands:

$$\mathcal{Q}_B = \left\{q \in [(1-\alpha)q^0, (1+\alpha)q^0]\colon \sum_{i\in\Omega}(q_i - q_i^0) \leqslant \beta\sum_{i\in\Omega}[(1+\alpha)q_i^0 - q_i^0]\ \forall \Omega \in \{NE, NW, SW, SE\}\right\}.$$

The support $\mathcal{Q}_B$ stipulates that the customer demands deviate by at most $\alpha \cdot 100\%$ from their nominal values q^0 specified in the benchmark problems. Moreover, the cumulative demand in each quadrant may not exceed its nominal value by more than $\beta \cdot 100\%$. We specify values for the two scalar parameters $\alpha, \beta \in [0, 1]$ later. Note that $\mathcal{Q}_B$ generalizes the rectangular supports studied in the literature (which correspond to $\beta = 1$) and the deterministic CVRP (which is recovered by setting $\alpha = 0$). We also consider the following factor model support for the customer demands:

$$\mathcal{Q}_F = \left\{q \in [(1-\alpha)q^0, (1+\alpha)q^0]\colon \exists \xi \in \Xi, \text{ such that } q_i = \left(1 + \alpha\sum_{f=1}^{4}\gamma_{if}\xi_f\right)q_i^0\ \forall i \in V_C\right\},$$

where

$$\Xi = \{\xi \in \mathbb{R}^4\colon \xi \in [-\mathrm{e}, +\mathrm{e}],\ \mathrm{e}^\top\xi \in [-4\beta, +4\beta]\}.$$

Similar to the case of $\mathcal{Q}_B$, the support $\mathcal{Q}_F$ stipulates that the customer demands deviate by at most $\alpha \cdot 100\%$ from their nominal values q^0. The demand of customer i is modeled as a convex combination of $f = 1, \ldots, 4$ factors that can be interpreted as quadrant demands. The weights γ_{if} of this convex combination reflect the relative proximity of customer i to each quadrant. More precisely, we set $\gamma_{if} = \rho_{if}/\sum_{f'=1}^{4}\rho_{if'}$, where ρ_{if} denotes the inverse distance between customer i and the centroid of quadrant f. The cumulative demand over all quadrants is allowed to deviate from its nominal value by at most $\beta \cdot 100\%$. Similar to $\mathcal{Q}_B$, we retrieve a rectangular support if we set $\beta = 1$ and the deterministic CVRP if we choose $\alpha = 0$.

The CVRP benchmark problems are characterized by a high utilization under the nominal demands q^0, that is, the unused capacity of each vehicle is small in every feasible set of routes. This is desirable for computational studies of $\mathcal{CVRP}$ since instances with small slack capacities are particularly difficult to solve. However, a high utilization is undesirable for computational studies of $\mathcal{RVRP}$ because the instances become infeasible even for modest demand uncertainty. To verify this statement, we solve the feasibility problem associated with $\mathcal{RVRP}$ for the selected benchmark problems. The feasibility problem can be cast as a robust bin packing problem (RBPP), which concerns the assignment of the customers to m bins of size Q such that the weights q_i of the customers $i \in V_C$ assigned to each bin do not exceed Q for any weight realization $q \in \mathcal{Q}$. Because the RBPP ignores the routing aspect of $\mathcal{RVRP}$, it can be solved much faster. About 50% of the instances are RBPP-infeasible for $(\alpha, \beta) = (0.1, 0.5)$, and this fraction rises to 95% for $(\alpha, \beta) = (0.2, 1.0)$. To alleviate this problem, we increase the vehicle capacities Q in each benchmark problem by 20%, which suffices to guarantee robust feasibility of every instance for uncertainty levels $\alpha \leqslant 0.2$.

Table 1 compares the performance of the seven proposed $\mathcal{RVRP}$ formulations from §4, with and without the robust RCI cut separation described in §5. We fix the uncertainty level to $(\alpha, \beta) = (0.1, 0.5)$, and we select

Table 1. Computational comparison of the $\mathcal{RVRP}$ formulations.

	Proven optimal			Residual gap†		
Formulation	#	Avg. nodes	Avg. time (sec)	#	Avg. gap (%)	No incumbent† #
			With robust RCI cut separation			
(2VF)	34	35,763	1,362	6	2.69	0
(MTZ)	15	102	5,891	12	9.25	13
(PRE)	13	85	1,377	14	6.51	13
(1CF)	12	76	9,423	8	10.53	20
(2CF)	16	110	6,970	7	6.94	17
(VA1)	27	13,821	3,890	13	3.01	0
(VA2)	23	17,342	1,809	17	3.01	0
			Without robust RCI cut separation			
(2VF)	6	2,166,633	258	32	39.02	2
(MTZ)	2	18,429	5,045	20	72.21	18
(PRE)	4	12,075	2,403	32	48.71	4
(1CF)	1	21,869	30,676	17	22.84	22
(2CF)	3	1,586	1,531	14	21.72	23
(VA1)	4	36,242	52	26	50.80	10
(VA2)	4	64,915	290	35	43.45	1

Note. To gurantee correctness, we always add those cuts in (2VF) and (VA1) that are required to eliminate infeasible binary node solutions.

†After 10 h CPU.

20 benchmark problems for each of the supports $\mathcal{Q}_B$ and $\mathcal{Q}_F$, resulting in a total of 40 instances. The results from §4 show that all seven formulations are necessary and sufficient for the robust feasibility of a set of routes $\mathbf{R}(x)$. This implies that the same route sets are feasible in each formulation. Moreover, because all formulations share the same objective function, they are optimized by the same route sets, which correspond to solutions that are equally robust. However, the table points out that some of the problem formulations are advantageous from a computational perspective. In particular, the table shows that the robust RCI cuts are crucial for solving any of the $\mathcal{RVRP}$ formulations. Moreover, we see that Formulations (2VF) and (VA1) can be solved most efficiently. Note that these are the two formulations that rely on robust RCI cuts to guarantee robust feasibility. In contrast, the other formulations possess tighter relaxations than (2VF) and (VA1), and they do not require robust RCI cut separation. However, these advantages come at a cost. Functional (wait-and-see) decisions, such as u in the MTZ formulation and f in the commodity flow formulations, require the introduction of auxiliary variables and dense constraints in the robust counterparts. As a result, the relaxations that are solved in the branch-and-bound nodes of Formulations (MTZ), (PRE), (1CF), and (2CF) require more computation time on average, and Table 1 indicates that this trade-off does not seem to pay off for $\mathcal{RVRP}$. We remark that there is no noticeable performance difference between the two supports $\mathcal{Q}_B$ and $\mathcal{Q}_F$.

Section EC.4 in the e-companion documents the performance of Formulation (2VF) with robust RCI cuts over 90 benchmark problems and the two supports $\mathcal{Q}_B$ and $\mathcal{Q}_F$, using the uncertainty level $(\alpha, \beta) = (0.1, 0.5)$. Most of the instances with fewer than 50 nodes can be solved to optimality. Out of those that were not solved to optimality, the optimality gap is consistently below 5%. The average gap over the instances with 50 nodes or more is 6.5%.

We remind the reader that the set-valued mapping $\mathcal{R}$ of feasible solutions to $\mathcal{RVRP}$ is monotonous, that is, $\mathcal{R}(\mathcal{Q}) \subseteq \mathcal{R}(\mathcal{Q}')$ if the support $\mathcal{Q}$ is a superset of the support $\mathcal{Q}'$. In particular, we have $\mathcal{R}(\mathcal{Q}) \subseteq \mathcal{R}(q^0)$ for any support $\mathcal{Q}$ that contains the nominal demands q^0. In the context of $\mathcal{Q}_B$ and $\mathcal{Q}_F$, this implies that any robust feasible set of routes is also feasible in $\mathcal{CVRP}$, and any robust feasible set of routes for the uncertainty level (α, β) is also robust feasible for the uncertainty levels (α', β'), $\alpha' \leqslant \alpha$ and $\beta' \leqslant \beta$. At the same time, a higher degree of robustness could incur higher costs, that is, we have to pay the "price of robustness" (Bertsimas and Sim 2004). This is illustrated in Tables 2 and 3, where the increase in transportation costs over the deterministic case is plotted for a number of uncertainty levels (α, β). In these tables, $\alpha = 0$ corresponds to the optimal solution to $\mathcal{CVRP}$, independent of the choice of β, and $\beta = 1$ corresponds to optimal solution over the rectangular support $[(1-\alpha)q^0, (1+\alpha)q^0]$. Note that the optimal objective values for rectangular and nonrectangular supports differ in general, which is consistent with our findings in §3.

7. Conclusions

This paper studied the robust capacitated vehicle routing problem (CVRP), in which the decision maker selects minimum cost vehicle routes that remain feasible for all realizations of the uncertain customer demands. We presented conditions under which this problem reduces to a deterministic CVRP, and we derived formulations for the generic case where the robust and the deterministic CVRP suggest different routes. Robust rounded capacity inequalities were proposed, and it was demonstrated that they can be used

Table 2. Impact of α and β in the supports $\mathcal{Q}_B$ (left) and $\mathcal{Q}_F$ (right) on the optimal solution of instance A-n32-k5.

$\alpha\backslash\beta$	0.00	0.25	0.50	0.75	1.00	$\alpha\backslash\beta$	0.00	0.25	0.50	0.75	1.00
0.00	740	740	740	740	740	0.00	740	740	740	740	740
0.05	740	742	748	748	748	0.05	740	742	742	742	748
0.10	742	748	748	756	756	0.10	742	748	748	756	756
0.15	742	756	761	761	761	0.15	748	748	756	756	761
0.20	756	761	784	784	784	0.20	748	756	761	761	784

Table 3. Impact of α and β in the supports $\mathcal{Q}_B$ (left) and $\mathcal{Q}_F$ (right) in terms of increase in transportation costs relative to the parameter setting $(\alpha, \beta) = (0, 0)$.

$\alpha\backslash\beta$	0.00 (%)	0.25 (%)	0.50 (%)	0.75 (%)	1.00 (%)	$\alpha\backslash\beta$	0.00 (%)	0.25 (%)	0.50 (%)	0.75 (%)	1.00 (%)
0.00	0.0	0.0	0.0	0.0	0.0	0.00	0.0	0.0	0.0	0.0	0.0
0.05	1.1	1.2	1.6	1.8	1.8	0.05	0.7	1.1	1.3	1.6	1.8
0.10	1.6	2.4	3.0	3.4	3.5	0.10	1.4	2.1	2.5	3.0	3.4
0.15	2.6	3.5	4.4	4.7	4.8	0.15	2.0	2.7	3.4	3.9	4.6
0.20	3.1	5.0	6.0	7.4	7.5	0.20	2.6	3.7	4.4	5.7	7.5

Note. The table reports the average across the 20 benchmark instances from Table 1.

to expedite the solution of the robust CVRP. Finally, the e-companion to this paper shows how the robust CVRP relates to the chance-constrained CVRP.

Several avenues for future work can be identified. First, it would be promising to derive the robust counterparts of other classes of cuts, such as the comb inequalities or the framed capacity constraints; see Chapter 3 in Toth and Vigo (2002). Computational experience with the deterministic CVRP suggests that this might significantly improve the computation times of branch-and-cut algorithms. Second, the formulations from §4 can be generalized to account for uncertain transportation costs, as well as uncertain travel and service times with time windows and maximum route durations. The extension to uncertain transportation costs can be achieved through an epigraph formulation; see Ben-Tal et al. (2009). The consideration of uncertain travel and service times is more involved because it requires an adaptation of the arguments from §4 to extended deterministic models that include time windows and route duration constraints (Baldacci et al. 2010, Cordeau et al. 2007, Toth and Vigo 2002). Finally, it would be of interest to extend our models to other types of uncertainty sets and more sophisticated recourse decisions, such as detours to the depot whenever a customer demand exceeds the remaining vehicle load.

Supplemental Material

Supplemental material to this paper is available at http://dx.doi.org/10.1287/opre.1120.1136.

Acknowledgments

The authors gratefully acknowledge partial support from the National Science Foundation [CMMI-0856021] and the Engineering and Physical Sciences Research Council [EP/H0204554/1].

References

Adelman D (2004) A price-directed approach to stochastic inventory/routing. *Oper. Res.* 52(4):499–514.

Ak A, Erera AL (2007) A paired-vehicle recourse strategy for the vehicle-routing problem with stochastic demands. *Transportation Sci.* 41(2):222–237.

Augerat P, Belenguer JM, Benavent E, A. Corberán, Naddef D (1998) Separating capacity constraints in the CVRP using tabu search. *Eur. J. Oper. Res.* 106(2–3):546–557.

Baldacci R, Hadjiconstantinou E, Mingozzi A (2004) An exact algorithm for the capacitated vehicle routing problem based on a two-commodity network flow formulation. *Oper. Res.* 52(5):723–738.

Baldacci R, Mingozzi A, Roberti R (2012) Recent exact algorithms for solving the vehicle routing problem under capacity and time window constraints. *Eur. J. Oper. Res.* 218(1):1–6.

Baldacci R, Toth P, Vigo D (2010) Exact algorithms for routing problems under vehicle capacity constraints. *Ann. Oper. Res.* 175(1):213–245.

Ben-Tal A, El Ghaoui L, Nemirovski A (2009) *Robust Optimization* (Princeton University Press, Princeton, NJ).

Ben-Tal A, Chung BD, Mandala SR, Yao T (2011) Robust optimization for emergency logistics planning: Risk mitigation in humanitarian relief supply chains. *Transportation Res. B* 45(8):1177–1189.

Ben-Tal A, Goryashko A, Guslitzer E, Nemirovski A (2004) Adjustable robust solutions of uncertain linear programs. *Math. Programming Ser. A* 99(2):351–376.

Bent RW, Van Hentenryck P (2004) Scenario-based planning for partially dynamic vehicle routing with stochastic customers. *Oper. Res.* 52(6):977–987.

Bertsimas D, Sim M (2003) Robust discrete optimization and network flows. *Math. Programming Ser. B* 98(1–3):49–71.

Bertsimas D, Sim M (2004) The price of robustness. *Oper. Res.* 52(1):35–53.

Bertsimas D, Simchi-Levi D (1996) A new generation of vehicle routing research: Robust algorithms, addressing uncertainty. *Oper. Res.* 44(2):286–304.

Bertsimas D, Brown DB, Caramanis C (2011) Theory and applications of robust optimization. *SIAM Rev.* 53(3):464–501.

Bertsimas D, Iancu DA, Parrilo PA (2010) Optimality of affine policies in multistage robust optimization. *Math. Oper. Res.* 35(2):363–394.

Birge JR, Louveaux F (2011) *Introduction to Stochastic Programming*, 2nd ed. (Springer, New York).

Ceria S, Stubbs RA (2006) Incorporating estimation errors into portfolio selection: Robust portfolio construction. Axioma Research Paper 3, Axioma, Inc., New York

Chen W, Sim M, Sun J, Teo C-P (2010) From CVaR to uncertainty set: Implications in joint chance-constrained optimization. *Oper. Res.* 58(2):470–485.

Chen X, Sim M, Sun P, Zhang J (2008) A linear-decision based approximation approach to stochastic programming. *Oper. Res.* 56(2):344–357.

Christiansen CH, Lysgaard J, Wøhlk S (2009) A branch-and-price algorithm for the capacitated arc routing problem with stochastic demands. *Oper. Res. Lett.* 37(6):392–398.

Cordeau J-F, Laporte G, Savelsbergh MWP, Vigo D (2007) Vehicle routing. Barnhart C, Laporte G, eds. *Transportation*. Handbooks in Operations Research and Management Science, Vol. 14, Chap. 6 (North-Holland, Amsterdam), 367–428.

Delage E, Ye Y (2010) Distributionally robust optimization under moment uncertainty with application to data-driven problems. *Oper. Res.* 58(3):596–612.

Desrochers M, Laporte G (1991) Improvements and extensions to the Miller-Tucker-Zemlin subtour elimination constraints. *Oper. Res. Lett.* 10(1):27–36.

Díaz BD (2011) The VRP Web. Accessed May 22, 2011, http://neo.lcc.uma.es/radi-aeb/WebVRP.

Dror M, Trudeau P (1986) Stochastic vehicle routing with modified savings algorithm. *Eur. J. Oper. Res.* 23(2):228–235.

Dror M, Laporte G, Louveaux F (1993) Vehicle routing with stochastic demands and restricted failures. *Math. Methods Oper. Res.* 37(3):273–283.

Dror M, Laporte G, Trudeau P (1989) Vehicle routing with stochastic demands: Properties and solution frameworks. *Transportation Sci.* 23(3):166–176.

El Ghaoui L, Oks M, Oustry F (2003) Worst-case value-at-risk and robust portfolio optimization: A conic programming approach. *Oper. Res.* 51(4):543–556.

Erera AL, Morales JC, Savelsbergh MWP (2009) Robust optimization for empty repositioning problems. *Oper. Res.* 57(2):468–483.

Erera AL, Morales JC, Savelsbergh MWP (2010) The vehicle routing problem with stochastic demand and duration constraints. *Transportation Sci.* 44(4):474–492.

Gendreau M, Laporte G, Séguin R (1995) An exact algorithm for the vehicle routing problem with stochastic demands and customers. *Transportation Sci.* 29(2):143–155.

Gendreau M, Laporte G, Séguin R (1996) Stochastic vehicle routing. *Eur. J. Oper. Res.* 88(1):3–12.

Golden B, Raghavan S, Wasil E, eds. (2008) *The Vehicle Routing Problem: Latest Advances and New Challenges* (Springer, New York).

Golden BL, Yee JR (1979) A framework for probabilistic vehicle routing. *AIIE Trans.* 11(2):109–112.

Golden BL, Magnanti TL, Nguyen HQ (1977) Implementing vehicle routing algorithms. *Networks* 7(2):113–148.

Gounaris CE, Repoussis PP, Tarantilis CD, Floudas CA (2011) A hybrid branch-and-cut approach for the capacitated vehicle routing problem. Pistikopoulos EN, Georgiadis MC, Kokossis AC, eds. *Computer-Aided Chemical Engineering: 21st European Symposium on Computer Aided Process Engineering* (Elsevier, Amsterdam), 507–511.

Gouveia L (1995) A result on projection for the vehicle routing problem. *J. Oper. Res. Soc.* 85(3):610–624.

Groër C, Golden B, Wasil E (2009) The consistent vehicle routing problem. *Manufacturing Service Oper. Management* 11(4):630–643.

IBM Corporation (2009) ILOG CPLEX Optimization Suite 12. IBM, Armonk, NY.

Jaillet P (1988) A priori solution of a traveling salesman problem in which a random subset of the customers are visited. *Oper. Res.* 36(6):929–936.

Janak SL, Lin X, Floudas CA (2007) A new robust optimization approach for scheduling under uncertainty: II. Uncertainty with known probability distribution. *Comput. Chemical Engrg.* 31(3):171–195.

Kara I, Laporte G, Bektas T (2004) A note on the lifted Miller-Tucker-Zemlin subtour elimination constraints for the capacitated vehicle routing problem. *Eur. J. Oper. Res.* 158(3):793–795.

Kenyon AS, Morton DP (2003) Stochastic vehicle routing with random travel times. *Transportation Sci.* 37(1):69–82.

Kleywegt AJ, Nori VS, Savelsbergh MWP (2004) Dynamic programming approximations for a stochastic inventory routing problem. *Transportation Sci.* 38(1):42–70.

Kuhn D, Wiesemann W, Georghiou A (2011) Primal and dual linear decision rules in stochastic and robust optimization. *Math. Programming Ser. A* 130(1):177–209.

Kulkarni RV, Bhave PR (1985) Integer programming formulations of vehicle routing problems. *Eur. J. Oper. Res.* 20(1):58–67.

Laporte G (2009) Fifty years of vehicle routing. *Transportation Sci.* 43(4):408–416.

Laporte G, Louveaux F, Mercure H (1992) The vehicle routing problem with stochastic travel times. *Transportation Sci.* 26(3):161–170.

Laporte G, Louveaux F, Van Hamme L (2002) An integer *L*-shaped algorithm for the capacitated vehicle routing problem with stochastic demands. *Oper. Res.* 50(3):415–423.

Laporte G, Nobert Y, Desrochers M (1985) Optimal routing under capacity and distance restrictions. *Oper. Res.* 33(5):1050–1073.

Li Z, Ding R, Floudas CA (2011) A comparative theoretical and computational study on robust counterpart optimization: I. Robust linear optimization and robust mixed integer linear optimization. *Indust. Engrg. Chemical Res.* 50(18):10567–10603.

Lin X, Janak SL, Floudas CA (2004) A new robust optimization approach for scheduling under uncertainty: I. Bounded uncertainty. *Comput. Chemical Engrg.* 28(6–7):1069–1085.

Lysgaard J, Letchford AN, Eglese RW (2004) A new branch-and-cut algorithm for the capacitated vehicle routing problem. *Math. Programming Ser. A* 100(2):423–445.

Ordóñez F (2010) Robust vehicle routing. Hasenbein JJ, ed. *INFORMS TutORials in Operations Research*, Chap. 7 (INFORMS, Hanover MD), 153–178.

Prékopa A (1995) *Stochastic Programming* (Kluwer Academic Publishers, Dordrecht, The Netherlands).

Ralphs TK, Kopman L, Pulleyback WR, Trotter LE (2003) On the capacitated vehicle routing problem. *Math. Programming Ser. B* 94(2–3):343–359.

Secomandi N, Margot F (2009) Reoptimization approaches for the vehicle-routing problem with stochastic demands. *Oper. Res.* 57(1):214–230.

Shapiro A, Dentcheva D, Ruszczyński A (2009) *Lectures on Stochastic Programming: Modeling and Theory* MPS-SIAM Series on Optimiation (SIAM, Philadelphia).

Shen Z, Dessouky MM, Ordóñez F (2009) A two-stage vehicle routing model for large-scale bioterrorism emergencies. *Networks* 54(4):255–269.

Smith SL, Pavone M, Bullo F, Frazzoli E (2010) Dynamic vehicle routing with priority classes of stochastic demands. *SIAM J. Control Optim.* 48(5):3224–3245.

Stewart WR, Golden BL (1983) Stochastic vehicle routing: A comprehensive approach. *Eur. J. Oper. Res.* 14(4):371–385.

Sungur I, Ordóñez F, Dessouky MM (2008) A robust optimization approach for the capacitated vehicle routing problem with demand uncertainty. *IIE Trans.* 40(5):509–523.

Sungur I, Ren Y, Ordóñez F, Dessouky MM, Zhong H (2010) A model and algorithm for the courier delivery problem with uncertainty. *Transportation Sci.* 44(2):193–205.

Tarantilis CD, Stavropoulou F, Repoussis PP (2012) A template-based tabu search algorithm for the consistent vehicle routing problem. *Expert Systems Appl.* 39(4):4233–4239.

Toth P, Vigo D, eds. (2002) *The Vehicle Routing Problem*. SIAM Monographs on Discrete Mathematics and Applications (SIAM, Philadelphia).

Verderame PM, Floudas CA (2009) Operational planning of large-scale industrial batch plants under demand due date and amount uncertainty. I. Robust optimization framework. *Indust. Engrg. Chemistry Res.* 48(15):7214–7231.

Verderame PM, Floudas CA (2011) Multisite planning under demand and transportation time uncertainty: Robust optimization and conditional value-at-risk frameworks. *Indust. Engrg. Chemistry Res.* 50(9):4959–4982.

Verderame PM, Elia JA, Li J, Floudas CA (2010) Planning and scheduling under uncertainty: A review across multiple sectors. *Indust. Engrg. Chemistry Res.* 49(9):3993–4017.

Yang W-H, Mathur K, Ballou RH (2000) Stochastic vehicle routing problem with restocking. *Transportation Sci.* 34(1):99–112.

Chrysanthos E. Gounaris is a postdoctoral research associate at the Computer-Aided Systems Laboratory at Princeton

University. His research involves the development of mathematical optimization-based methodologies to address problems in the areas of process operations, process systems engineering, materials science, and computational chemistry.

Wolfram Wiesemann is an assistant professor at the Imperial College Business School. His current research focuses on the development of tractable computational methods for the solution of stochastic and robust optimization problems, as well as applications in energy systems and operations management.

Christodoulos A. Floudas is the Stephen C. Macaleer '63 Professor in Engineering and Applied Science, professor of chemical and biological engineering, associated faculty in the Program of Applied and Computational Mathematics, associated faculty in the Department of Operations Research and Financial Engineering, and faculty in the Center for Quantitative Biology at Princeton University. He is a member of the U.S. National Academy of Engineering. His research interests lie at the interface of chemical engineering, applied mathematics, and operations research, with principal areas of focus including chemical process synthesis and design, process control and operations, discrete-continuous nonlinear optimization, local and global optimization, and computational chemistry and molecular biology.